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SUPERSEDING
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MILITARY SPECIFICATION

BALLOON, METEOROLOGICAL ML-537( )/UM

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of sounding balloon that is used with meteorological equipment and is capable of ascents to high altitudes during both daytime and nighttime hours (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-B-14417	- Balloons, Meteorological, Pilot, Ceiling and Sounding, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6660

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STANDARDS

FEDERAL

FED-STD-601 - Rubber, Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein.

NATIONAL BUREAU OF STANDARDS

JOURNAL OF RESEARCH, NATIONAL BUREAU OF STANDARDS, Volume 53, No. 6, December 1954, Research Paper No. 2557, "Aerological Sounding Balloons."

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish sample units for first article inspection and approval (see 4.3 and 6.2).

3.2 Construction. The balloons shall be constructed in accordance with the requirements herein, and shall be capable of passing the tests of section 4.

3.2.1 Material. The balloons shall be fabricated of neoprene latex. The material shall be of uniform texture and shall be free from impurities and pinholes. The balloons shall be post-plasticized to provide freeze resistance for nighttime flights. In addition to the requirements of this specification, the following shall apply:

a. The requirements of MIL-P-11268, including the selection requirement therein.

b. Recycled virgin and reclaimed materials. Without jeopardizing the intended use of equipment furnished to this specification, selection of materials will be governed by the following:

(1) There is no exclusion to the use of recovered materials.

(2) There is no requirement that equipment be manufactured from virgin materials.

(3) Reclaimed materials are required to the maximum extent possible.

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3.2.2 Physical properties. The material used in fabrication of the balloon envelopes shall meet the requirements of either (a) column A (using test methods of FED-STD-601), or (b) column B (using National Bureau of Standards patch test method) shown as follows:

Requirement	Prior to aging 1/		After aging 1/		Cold temperature 4/	
	A	B	A	B	A	B
Elongation at break (min)	750%	7.00	650%	6.50	500% 2/	4.50
Tensile strength (psi, min)	1100	1100	1100	1100		
Tensile stress (modulus) 3/ (psi, min)	200	200	200	200		
(psi, max)			500	500		

1/ At 20°C (68°F).

2/ With no sign of cold flow.

3/ At 600% elongation.

4/ At -70°C (-94°F).

3.2.3 Neck. The balloon shall have a tubular neck of uniformly round cross-sectional area. The neck shall be molded integrally with the body of the balloon or shall be securely attached thereto. The neck shall have the following dimensions and characteristics:

3.2.3.1 Length. The tubular portion of the neck shall have a minimum length of 4.5 inches (114.3mm).

3.2.3.2 Thickness. The neck shall have a minimum wall thickness of 0.025 inch (.64mm) and a maximum wall thickness of 0.075 inch (1.9mm).

3.2.3.3 Diameter. The inside diameter of the neck shall be 1 ±.125 inch (25.4 ±3.2mm).

3.2.3.4 Neck strength. The balloon neck shall be capable of withstanding a load (force) of 40 pounds (18.1kg) for one minute when tested according to 4.7.

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3.2.4 Ozone resistance. The balloon shall be capable of withstanding exposure to a concentration of ozone of 150 parts per one hundred million when tested in accordance with 4.10.

3.2.5 Bursting requirements. The balloon shall reach a minimum volume of 8500 cubic feet (240.7 m^3) or a minimum diameter of 25.3 feet (7.7m) at burst.

3.2.6 Balloon length. The maximum overall length of the balloon shall be not greater than 120 inches (304.8cm).

3.3 Flight requirements. The balloons, inflated with hydrogen, shall have a total lift of 3900 ±100 grams (137.6 ±3.5 ozs) and when released shall carry a load of 1300 ±50 grams (45.9 ±1.8 ozs) exclusive of the weight of the balloon. Under these conditions, 80 percent of the balloons shall have a bursting altitude of 100,000 feet (30,480m) or more and a rate of rise equal to or in excess of 1000 feet (304.8m) per minute. The balloons shall not exhibit a tendency to level off at any altitude during flight (see 4.12 and 6.6).

3.4 Color of balloon. The balloon shall be the color of natural cured neoprene with no pigment added. However, antioxidants and antiozonants added to the balloon material to improve aging and ozone resistance of the balloon film may result in a dark appearance which in no way adversely affects the performance of the balloon.

3.5 Balloon number. First article samples of balloons shall be numbered consecutively beginning with number "1." Balloons from production shall be numbered consecutively beginning with the number following that of the last first article sample. The number shall be stamped on the individual balloon container.

3.6 Balloon quality. Balloons shall not be more than 120 days old when submitted for inspection and acceptance. Finished balloons shall be of uniform texture and shall contain no patches. The material shall contain a suitable antioxidant to preserve the balloons in a serviceable condition. Balloons shall be cured by a process that shall not materially impair their elastic properties. The date of manufacture shall be stamped on the individual balloon container.

3.7 Dusting. Balloons shall be dusted evenly inside and outside with a dry powdered material which will effectively prevent cohesion and adhesion. This powder shall not react with hydrogen in an unfavorable manner or be injurious in any way to personnel handling the balloons.

3.8 Balloon containers. Balloons shall be packaged in individual plastic bags, fabricated of a material that will not affect the neoprene balloon film, and the bags shall be sealed to produce an air-proof barrier. A simple method of opening the bag, such as a rip strip or corrugated tear seal, shall be provided. The plastic bag containing the balloon shall be placed within an individual container designed to provide adequate protection for the balloon. The balloons shall not be packaged so tightly that structural weakness of the balloon film can develop during storage.

3.8.1 Inclosure. The following instructions shall be printed legibly on an appropriately sized, single sheet of paper, one of which shall be inclosed in each individual balloon container:

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IMPORTANT NOTICE

Balloon film has an initial thickness of approximately 0.0035 inch (.0089cm) and a thickness at burst of approximately 0.0001 inch (.0025mm). It is extremely fragile material and a slight surface abrasion caused by handling can impair performance seriously. To avoid damage, prepare balloon for flight as follows:

a. Remove the plastic bag containing the balloon from the container. Open the bag completely using the rip strip or tear seal provided and lift the balloon from the bag, supporting it from below. Never pull the balloon from the bag.

b. Inspect the balloon for holes. Reject any balloon having holes in the film. Holes in the balloon film shall be the only basis for rejection. Discoloration does not affect balloon performance.

c. Omit preconditioning if the balloon is less than one year old (date of manufacture is stamped on container) and has not been subjected to low temperatures, 32°F (0 °C) or below.

d. If the balloon has been subjected to low temperatures, 32°F (0 °C) or below, remove the balloon from its container and bag, unroll it gently and maintain at a temperature of approximately 70°F (21.1°C) for 24 hours. Balloons which have been exposed to such conditions and balloons which are more than one year old must be preconditioned prior to flight. Follow the instructions in the technical bulletin as issued for your guidance. If you have no other instructions, immerse the balloon in boiling water for about 5 minutes. Inflate and release the balloon as soon as possible thereafter. If ambient temperature at ground level is above 40°F (4.4°C), the balloon can be released while still wet. If the temperature is below 40°F (4.4°C), dry the balloon in a warm area prior to release.

e. Always exercise extreme care during all phases of handling the balloon.

3.9 Marking.

3.9.1 Individual container. Each individual container shall have stamped thereon the nomenclature of the balloon to be contained therein, the balloon weight in grams, the balloon number, and the date of manufacture.

3.9.2 Outer container. The following instructions shall be printed on the outside of each outer or shipping container.

STORAGE

Store at a temperature of not less than 40°F (4.4°C) and not more than 80°F (26.7°C). Store away from heat sources such as sunlight, steam pipes, heating ducts, or other heat radiating devices and also away from operating electrical equipment and ultra-violet radiation devices.

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3.10 Workmanship. Balloons shall be manufactured and assembled in accordance with the following paragraphs herein:

3.2.1	Material	3.7	Dusting
3.4	Color	3.8	Containers
3.5	Balloon number	3.8.1	Inclosures
3.6	Quality	3.9	Marking

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3) (does not include packaging).

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspection:

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection for packaging (see 4.14).

4.3 First article. Unless otherwise specified in the contract, the first article inspection, with the exception of the flight test, shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish samples as required by contract for group A, group B and group C testing.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in tables I, II, and III. The 24 balloons selected for flight tests shall not be subjected to the physical properties tests and vice-versa. Since only 20 sets of flight test data shall be obtained, the remaining 4 balloons shall be used only in the event of radiosonde failure or if a balloon bursts on the ground. There shall be no more than 4 balloon failures in this test.

4.3.3 First article data. The first article test plan and test report shall be as required in the contract.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable documents as part of the inspection required by this specification, and the inspection requirements specified in the contract:

Item	Requirement paragraph
Parts, materials and processes	3.2.1
Marking	3.9
Workmanship	3.10

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.5.1 through 4.5.3 except for the flight test, which will be performed solely by the government. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspection in table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQLs indicated in table I.

TABLE I. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (% defective) Major	Minor
Visual and mechanical	3.10	4.4	1	4

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the government.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table II and to the special procedures for small-sample inspection of MIL-STD-105.

4.5.2.1 Group B sampling plan. The AQL shall be 6.5% defective and the inspection level shall be L-7 for normal and tightened inspection, and L-5 for reduced inspection. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

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4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in any order satisfactory to the government.

TABLE II. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)
Neck dimensions:			
Length	3.2.3.1	4.6.1	6.5 for the combined group
Thickness	3.2.3.2	4.6.2	
Diameter	3.2.3.3	4.6.3	
Balloon length	3.2.6	4.13	

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table III. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling of group C inspection. Sample balloons in each of the subgroups of table III shall be selected from each lot, without regard to their quality, from samples that have passed group A and B inspections. Balloons inspected at the start of the contract shall be selected from the first balloons produced. The number of samples and permissible failure rates for each subgroup shall be as follows:

a. Subgroup 1. Five balloons shall be selected, and one failure is permissible.

b. Subgroup 2. Twelve balloons shall be selected and two failures for any test are permissible, except that no more than a total of two balloons shall fail.

c. Subgroup 3. Five balloons shall be selected and one failure for any test is permissible, except that no more than one balloons shall fail.

4.5.3.2 Flight test data. Ten of the balloons selected for flight test shall be subjected to the tests of 4.12. In the event of radiosonde failure (see 4.12.7), a new balloon will be selected from the remaining two balloons and the flight test completed. Only ten sets of data shall be obtained.

4.5.3.3 Group C failures. Action required relative to group C failures shall be as specified in the contract.

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TABLE III. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Subgroup 1		
Burst test	3.2.5	4.11
Subgroup 2		
Flight test: Bursting altitude	3.3	4.12
Rate of rise		
Subgroup 3		
Neck strength	3.2.3.4	4.7
Ozone resistance	3.2.4	4.10
Physical properties:		
Prior to aging: 1/	3.2.2	
Tensile stress		4.8.3
Tensile strength		4.8.4
Elongation at break		4.8.5
After aging: 1/	3.2.2	4.8.6.2
Tensile stress		4.8.3
Tensile strength		4.8.4
Elongation at break		4.8.5
Cold temperature: 2/	3.2.2	
Elongation at break		4.9

1/ At 20°C (68°F)
2/ At -70°C (-94°F)

4.6 Neck dimensions tests. The dimensions of the balloon neck shall be gauged or measured as follows:

4.6.1 Length. The length of the tubular (approximately uniform cross-sectional area) portion of the neck shall be measured with a suitable scale.

4.6.2 Thickness. The thickness of the neck section shall be measured at 10 different places. The measurements may be made of a double thickness of the neck section provided that it is flattened and a true measure is obtained of only the double thickness.

4.6.3 Diameter. The inside diameter of the neck shall be measured by fitting the neck over rigid tubes or cylinders of circular cross-section whose outside diameters shall be .875 inch (22.2mm) and 1.125 inch (28.6mm). The neck shall fit over the .875 inch (22.2mm) tube or cylinder without stretching, but shall require stretching to fit over the 1.125 inch (28.6mm) tube or cylinder.

4.7 Neck strength tests. The five balloons selected in accordance with 4.5.3.1 for physical properties tests shall be subjected to the following test prior to being subjected to the physical properties tests.

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4.7.1 Purpose of test. This test is intended to demonstrate that the balloon neck and the balloon film adjacent to the neck have sufficient strength to withstand the normal stresses of an instrument flight.

4.7.2 Apparatus for test. The apparatus shall consist of a self-tightening jaw which grasps the neck of the balloon and to which a 40 pound (18.1kg) load, including the weight of the jaw, can be added. The balloon shall be wrapped around a smooth rod not less than 1 inch (25.4mm) in diameter so that no more than 3 inches (76.2mm) of the balloon itself is permitted to extend below the rod when the load is applied. The rod and balloon shall be suspended 10 feet (304.8cm) from the floor of the testing area.

4.7.3 Procedure for test. The jaw shall be fastened to the balloon neck after the balloon has been wrapped around the rod as described in 4.7.2 and the load shall be applied in increments of 5 pounds (2.3kg) until the total load is 40 pounds (18.1kg). Application of the load shall be accomplished in not less than one minute. The balloon and neck shall be allowed to extend freely during application of the load. The 40 pound (18.1kg) load shall be applied for a period of one minute without rupture of either the balloon or neck and without separation of the neck from the balloon if the neck is not an integral part of the balloon.

4.8 Physical properties tests. After subjection to the neck strength tests, the five balloons shall be utilized in conducting the physical properties tests. Elongation, tensile strength, and tensile stress (modulus) tests prior to aging shall be conducted concurrently on the same specimens.

a. When testing is to be in accordance with FED-STD-601, six specimens (two from near the neck, two from the balloon equator, and two from the end opposite the neck) each measuring 20 inches by 20 inches (508mm by 508mm) shall be cut from each balloon. Then, three dumbbell shaped specimens, using die III of method 4121, FED-STD-601, shall be cut from each of the 20 inch by 20 inch (508mm by 508mm) specimens (one from each of the three locations specified above) for testing prior to aging. The remaining three samples shall be utilized for testing after aging.

b. When testing in accordance with procedures contained in Research Paper 2557, Volume 53, No. 6 of the Journal of Research of the National Bureau of Standards, December 1954, the six specimen samples, two from each of the areas identified in 4.8a above, shall be cut to measure 30 inches by 30 inches (762mm by 762mm). Three segments measuring approximately 5 inches by 5 inches (127mm by 127mm) shall be cut from each of three of the 30 in. by 30 in. (762mm by 762mm) specimens (one from each location) for testing prior to aging and the remaining samples utilized for testing after aging.

4.8.1 Test apparatus. Test apparatus for the physical properties and cold temperature tests shall be as described in method 4111 of FED-STD-601. The procedure for using the patch method outlined in Research Paper 2557, Volume 53, No. 6 of the Journal of Research of the National Bureau of Standards is as follows:

a. Patch box. A cylindrical, hinged box similar to that shown in figure 1, which provides a means for clamping the specimen so that it has a

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circular exposed area having a diameter of 2.5 inches (63.5mm). The box shall include a valve which can be connected to a source of dry nitrogen.

b. The valve (connection) for the nitrogen supply shall be a cut-off type so that the patch box can be removed from the nitrogen supply for the ozone test of 4.10.

c. A flexible scale shall be so oriented that it passes over the pole (central point) and conforms to the shape of the patch during inflation for measurement of the increase in diameter of a 1 centimeter (0.4 inch) circle inscribed in the center of the patch.

d. Two freely moving flat-headed pistons which are pushed outward as the patch inflates for measurement of the major patch (bubble) diameter at 600% elongation and burst.

e. A means such as a manometer or bellows connected to a suitable electrical measuring device for measuring the pressure in the patch.

4.8.2 Test procedure. The test procedures utilizing FED-STD-601 are outlined in the applicable methods of that standard. The Bureau of Standards procedure is as indicated in this and the following subparagraphs for individual tests.

a. The patch shall be inflated at such a rate that rupture (burst) occurs in 17.5 ±2.5 minutes.

b. The median of the results from the tests of the nine specimens from each balloon shall be recorded as the final result for each test (tensile stress, tensile strength and elongation) for that balloon. If more than one such value (1 of 5 balloons) shall fall below the values specified in 3.2.2, the lot represented by the test specimens shall be considered as having failed the test and shall be cause for rejection of the lot.

4.8.3 Tensile stress (modulus). The tensile stress shall be measured at 600 percent elongation, and shall be determined according to method 4131 of FED-STD-601 or by following the procedure outlined in Research Paper 2557, Volume 53, No. 6, December 1954, of the Journal of Research of the National Bureau of Standards, except for temperature. The results of this test shall be determined and recorded prior to the tensile strength and elongation tests.

4.8.3.1 Test results. The method of determining the results is specified in FED-STD-601. Determination of results using the NBS method is by observing the pressure when the patch has reached the 600 percent elongation point (pistons and the equatorial radius of the patch at this point) and substituting these values for "P" and "R_e" in the following equations:

$$S_6 = \frac{1.35\ P\ R_e^{\ 2}}{at}$$

Where S_6 = Tensile stress at 600% elongation

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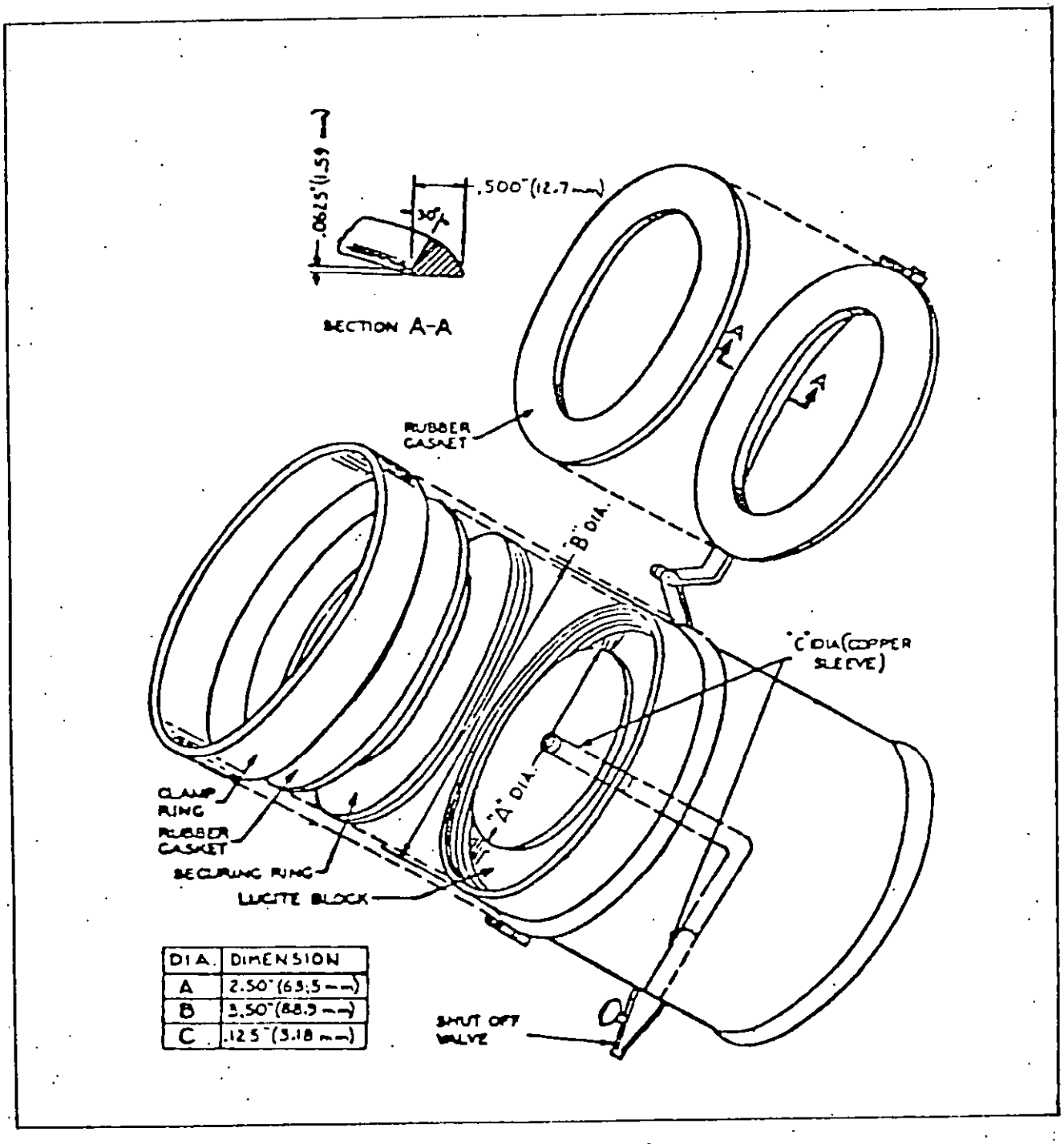


FIGURE 1. Patch box.

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P = Differential pressure across patch at 600% elongation

R_e = Equatorial radius of patch at 600% elongation

a = Radius of original exposed patch

t = Original thickness of film

This test shall be conducted both before and after aging of the balloon film using separate specimens.

4.8.4 Tensile strength. This test is defined as the pressure required to cause the test specimen to rupture or burst. The tensile strength shall be determined according to method 4111 of FED-STD-601 or by the procedure outlined in the National Bureau of Standards document cited in 4.8.3 except for temperature.

4.8.4.1 Test results. The method of determining test results is specified in FED-STD-601. The National Bureau of Standards procedure requires observing the pressure at rupture or burst of the patch and the spacing between the pistons (see 4.8.1d) after burst and substituting these values in the following equation:

$$S_B = \frac{1.35\ P\ R_e\ 2}{at}$$

Where S_B = Tensile strength

P = Differential pressure across patch at burst

R_e = Equatorial radius of patch at burst

a = Original radius of exposed patch

t = Original thickness of patch

This test shall be conducted before and after aging of the balloon film using separate specimens.

4.8.5 Elongation. Elongation at burst shall be determined according to method 4121 of FED-STD-601 or by the procedure outlined in the National Bureau of Standards document specified in 4.8.3 except for temperature. This test shall be conducted before and after aging of test samples, using separate specimens.

4.8.5.1 Test results. Test results shall be determined as specified in FED-STD-601. Determination by the National Bureau of Standards method is given as the ratio of the diameter of the inscribed circle at burst of the patch to that of the original diameter of the circle. Alternatively, elongation can be determined as follows:

$$E_B = \frac{2.60\ R}{a}$$

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Where E_B = Elongation at burst

R = Equatorial radius of patch at burst (measured by pistons)

a = Original radius of patch

4.8.6 Accelerated aging and associated tests. Accelerated aging of balloon film is intended to estimate the relative resistance of cured neoprene latex balloons to aging. No direct correlation between this accelerated aging and the natural life of neoprene balloons is given or implied. The procedure consists of subjecting test samples having previously determined physical properties to controlled elevated temperatures in air at atmospheric pressure. It does not include exposure to light. Changes in elongation at break, tensile strength, and tensile stress (see 4.8.3, 4.8.4, and 4.8.5) are the criteria of deterioration.

4.8.6.1 Apparatus. An air oven shall be used and shall meet the following requirements:

a. Provision shall be made to lay test samples between two layers of cheesecloth on a perforated platform.

b. The heating medium for the aging chamber shall be circulated air at atmospheric pressure.

c. The air shall be circulated by means of mechanical agitation. The air must not come in contact with any electric motor brush discharge or other source of ozone.

d. The source of heat shall be located outside of the aging chamber proper.

e. The temperature shall be controlled automatically.

f. A recording thermometer shall be used to record and maintain the temperature in the aging chamber.

g. Actual checks of the oven temperature shall be by means of a maximum reading thermometer.

4.8.6.2 Test procedure. Three of the test samples (20 in by 20 in or 30 in by 30 in) (508mm by 508mm or 762mm by 762mm) selected as in 4.8 are used for the accelerated aging of the balloon film. The oven shall be pre-heated to operating temperature, 100°C (212°F). Aging shall be conducted at 100°C ±5°C (212°F ±9°F) for a period of 8 hours. The test specimens (see 4.8) shall be cut after aging of the samples. All specimens shall be kept in a desiccator for a minimum of 24 hours after removal of the samples from the oven. Testing shall be conducted within 96 hours after removal of the samples from the oven. The tests shall be conducted for each one of the methods specified in 4.8.3, 4.8.4, and 4.8.5

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4.9 Elongation of -70°C (-94°F). This test is intended to demonstrate that the balloons are capable of expansion at temperatures encountered during flight without showing signs of freezing and that the film has sufficient elongation to permit the balloons to reach their required altitude. Three test specimens shall be cut from the test samples (one from each sample) selected for the physical properties tests before aging (see 4.8) dependent upon the test method used.

4.9.1 Test apparatus. A low temperature cabinet of one of the types described in method 5111 of FED-STD-601 (preferably the mechanically refrigerated type) shall be used. The cabinet shall have the following characteristics:

a. A refrigerated work space of approximately 2 ft by 2 ft by 2 ft (61cm by 61cm by 61cm).

b. The door shall be the front opening type and shall contain a 5 ply thermopane window or its equivalent to permit observation of the test sample during the test.

c. Shall be capable of maintaining a constant temperature of -70°C ±1°C (-94°F ±1.8°F).

d. The cabinet shall be equipped with a means of elongating a dumbbell specimen to "rupture" or with a means of inflating a patch according to the procedure used in the National Bureau of Standards method.

e. The rate of separation of the jaws holding the specimen shall be 0.75 in (19.1mm) per minute if a dumbell specimen is used. If the National Bureau of Standards patch method is used, the rate of inflation shall be such that rupture occurs in 17.5 ±2.5 minutes.

4.9.2 Test procedure. Specimens shall be conditioned at the test temperature for 10 ±1 minutes prior to beginning the test. Elongation is the only test conducted at this temperature.

4.9.3 Test results. The median value for the three specimens from each balloon shall be recorded as the final value. If more than one such median value (1 balloon) were lower than the value specified in 3.2.2, the lot represented by the specimens shall be considered as having failed the test and shall be cause for the rejection of the lot.

4.10 Ozone test. This test is designed to demonstrate that the balloons are capable of withstanding exposure to ozone concentrations normally encountered in the atmosphere. The test is conducted at 600% elongation of the test specimens.

4.10.1 Test apparatus. The test apparatus shall consist of a test chamber such as the GF Bush Associates Model B-2 Ozonator, or equal, having an approximate capacity of 0.6 cubic feet (.017m^3) with an ozone generating source outside the test chamber. It shall have the following characteristics:

a. The chamber shall have means for measuring ozone concentration. The method used for analysis shall be absorption of the ozone by a buffered solution of potassium iodide in water and measurement of the iodine liberated by titration with sodium thiosulphate.

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b. The test chamber shall be equipped with a removable rack so that the previously elongated specimens may be placed therein.

4.10.2 Test procedure. Six dumbbell specimens (two each from the three 20 inch by 20 inch (508mm by 508mm) samples) or three patches (one each from the three 30 inch by 30 inch (762mm by 762mm) samples) selected for testing of physical properties tests and unaged shall be used for this test. The specimens or patches shall be elongated 600% for the test. Provision shall be made for clamping the dumbbell specimens on the test rack in that condition or for elongating them on the rack when using the FED-STD-601 method. One inch (25.4mm) bench marks shall be placed on the dumbbell specimens to indicate elongation to 600%. Elongation, using the patch (NBS) method shall be as in 4.8. The sealed-off patch box with the patch elongated 600% shall be placed on the test rack. The patch shall remain elongated (inflated) 600% during the test.

4.10.2.1 Conditioning. The ozone chamber shall be allowed to reach equilibrium (steady or constant concentration), and the rack containing the specimens shall be placed in the chamber.

4.10.2.2 Concentration. The ozone concentration shall be measured (determined) for compliance with 3.2.4 twice during the period of any one test and shall be maintained constant during the test.

4.10.2.3 Length of test. The specimens shall be allowed to remain in the test chamber at a temperature of 20°C ±1°C (68°F ±1.8°F) for a period of 24 hours ±15 minutes.

4.10.3 Test results. A specimen shall be considered as having failed if it breaks during the test or if it shows signs of deep cracks or holes at any area other than where the bench marks are applied. A balloon shall be considered as having failed the test if more than two dumbell specimens or more than one patch from that balloon fails. Failure of more than one balloon to pass the test shall be cause for rejection of the lot represented by the test specimens or patches.

4.11 Burst test. Five of the balloons, selected in accordance with 4.5.3.1 shall be utilized for this test. The test is intended to demonstrate that the balloon is capable of expansion to a volume which will permit flights to 100,000 feet (30,480m) altitude when flown with the lift specified in 3.3. It is not a substitute for the flight test, and ability of the balloons to meet this test does not constitute a waiver of the flight requirements or test thereof.

4.11.1 Test apparatus. The following equipment and installation are required for this test:

a. A blower capable of delivering air at atmospheric pressure at a rate of not less than 80 cubic feet ($2.3m^3$) per minute.

b. An air volume meter reading in cubic feet or metric.

c. A nozzle of a size suitable for attachment to the balloon neck. The nozzle shall point vertically downward and shall be located at a height of not less than 30 feet (9.1m) above the testing floor.

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d. The above apparatus shall be installed in a building which provides unobstructed expansion of the balloon to a diameter of at least 30 feet (9.1m).

4.11.2 Test procedure. The balloon shall be fastened securely to the inflation nozzle so that no air can escape around the neck of the balloon. Inflation shall be conducted at a rate of not less than 80 cubic feet ($2.3m^3$) per minute, and shall be continued until the balloon bursts. The volume of the balloon shall be considered to be the same as the volume of air recorded by the air metering device. Alternatively, the horizontal (equatorial) diameter of the balloon at burst shall be measured by suitable means.

4.11.3 Test results. If more than one balloon fails to reach the minimum bursting volume (see 3.2.5), the lot shall be rejected.

4.12 Flight tests. The flight tests shall be conducted solely by the government at the location designated in the contract. The balloons shall be subjected to the following flight tests to determine compliance with the rate of rise and bursting altitude requirements of 3.3.

4.12.1 Conditioning for test. The balloons need not be preconditioned for flight tests.

4.12.2 Inflation. The balloons shall be inflated with hydrogen or helium. The lift for the balloons shall be as specified in 3.3. The bursting altitude requirement of 3.3 is applicable when using hydrogen. A bursting altitude of 96,000 feet (29,261m) is acceptable when using helium. The rate of rise requirement of 3.3 is applicable when using either hydrogen or helium.

4.12.3 Loading. The balloon shall be loaded with 1300 ±50 grams (45.9 ±1.8ozs) exclusive of the weight of the balloon. This load shall include a radiosonde, the load line, and a parachute. The parachute shall be attached to the neck of the balloon by means of a 100-pound (45.4kg) test nylon load line approximately 50 feet (15.3m) long. The radiosonde shall be suspended immediately below the parachute. The nylon load line shall be tied securely to the neck at a point approximately 2.5 inches (63.5mm) from the open end of the neck. The neck shall be doubled back upon itself and the folded neck shall be tied securely with a separate length of line at a point above that at which the load line is attached. It is important that this operation (tying) be carried out carefully since a slight leak due to incorrect assembly can affect the flight of the balloon in the same manner as a pinhole in the balloon. In addition, if the parachute and radiosonde become prematurely separated from the balloon, the monitoring equipment (radiosonde receptor) cannot distinguish this condition from a balloon burst.

4.12.4 Calibration of pressure element. The pressure element of the radiosonde shall be corrected (calibrated) for surface pressure. If a hypsometer type radiosonde is used, calibration prior to flight is not necessary.

4.12.5 Launching. The balloons shall be released in free flight at any time of day or night, except that both daytime and nighttime flights shall be accomplished during the test.

4.12.6 Record data. The bursting altitude and rate of rise shall be computed from the radiosonde (receptor) record.

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4.12.7 Radiosonde failure. If for any reason, not a fault of the balloon, a radiosonde record for determination of bursting altitude or rate of rise is not obtained, a new sample shall be selected, and the flight test shall be repeated using the substitute sample.

4.12.8 Balloon ground failure. If failure of the balloon due to a fault of the balloon on the ground prohibits its release, this condition shall be interpreted as a flight that failed the bursting altitude test. A substitute sample balloon shall then be released to determine the rate of rise.

4.13 Balloon length. The length of the balloon shall be measured for compliance with 3.2.6. This measurement shall be made after deflation of the balloon subsequent to visual and mechanical inspection.

4.14 Quality conformance inspection for packaging. Quality conformance inspection for packaging shall be in accordance with group A, group B and, when required (see 6.2d), the group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

4.15 Visual and mechanical inspection. Balloons and containers shall be inspected for defects listed in table IV.

TABLE IV. Classification of defects.

Classification	Defect
Major	Balloon contains foreign matter. Uneven texture. Uneven thickness. Evidence of patches. Date indicates balloon more than 120 days old. Balloons improperly powdered or not powdered. Balloon container too small. Notice sheet missing from inside of container. Storage notice sheet not on shipping container. Evidence of holes or cracks. Balloon contains non-removable pinches.
Minor	Balloon contains marked or rubbed areas. Uneven coloring. Balloon number incomplete or illegible. Containers too large.

5. PACKAGING

5.1 Packaging requirements. Packaging shall be in accordance with MIL-B-14417.

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6. NOTES

6.1 Intended use. Balloon, Meteorological MIL-537( )/UM is used for carrying radiosonde equipment to a specified height for obtaining meteorological (weather) information.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level A or level B preservation and packaging (see section 5).

c. When rough handling test is required (see section 4).

d. Marking and shipping of samples.

e. Place of final inspection (see 4.12).

f. Number of first articles to be submitted for approval (see 3.1).

6.3 Group C failures. Action required relative to group C failures shall be as specified in the contract.

6.4 Definitions. The following definitions are applicable to the flight requirements of 3.3:

a. Bursting altitude. Bursting altitude is the maximum height above sea level which the balloon reaches during flight while carrying its specified load.

b. Rate of rise. The rate of rise is the ratio of the maximum altitude above sea level that the balloon reaches to the time required to reach that altitude carrying its specified load.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - ER
Air Force - 99

Review activity:
Air Force - 80

Preparing activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-55087B BALLOON, METEOROLOGICAL ML-537( )/UM

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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